

Tilting of Radioactive Plaques After Initial Accurate Placement for Treatment of Uveal Melanoma

Arghavan Almony, MD; Sean Breit, MD; Hui Zhao, PhD; Jose Garcia-Ramirez, MS; David B. Mansur, MD; J. William Harbour, MD

Objective: To evaluate plaque movement as a potential factor in local failure using intraoperative ultrasonography at plaque insertion and removal.

Methods: Prospective study of 162 patients with uveal melanoma undergoing intraoperative B-scan ultrasonography at insertion and removal of iodine 125 plaques.

Results: Tilting of the posterior plaque edge more than 1.0 mm away from the sclera was detected in 15 patients (9%) at plaque insertion and in 85 patients (53%) at plaque removal ($P < .001$). Factors associated with tilt at plaque removal included male sex ($P = .009$), decreased tumor distance to the fovea and optic disc ($P < .001$ for both), notched plaque ($P = .001$), and episcleral hematoma ($P = .009$). Plaque

tilt caused a reduction greater than 10% in actual radiation dose to the tumor apex in 37 patients (23%). Local failure occurred in only 3 patients (2%), all of whom had tilt of 1.95 mm or greater at plaque removal.

Conclusions: Plaque tilt after initial accurate placement occurs frequently during brachytherapy for uveal melanomas and may represent an important cause of local treatment failure. Recognizing and counteracting the effects of plaque tilt may reduce the risk of local failure.

Trial Registration: clinicaltrials.gov Identifier: NCT00459849

Arch Ophthalmol. 2008;126(1):65-70

EPISCLERAL PLAQUE RADIOTHERAPY is a common treatment for uveal melanoma and results in local tumor control in most cases.¹ However, local failure has been reported in a substantial proportion of patients.²⁻⁶ Because local failure greatly increases the risk of metastatic death,⁷⁻⁹ identifying and correcting the causes of local treatment failure are of paramount importance. Poor plaque localization is an important contributing factor to local failure.^{10,11} With the aid of intraoperative ultrasonography, most malpositioned plaques can be identified and readjusted at the time of plaque insertion.¹⁰⁻¹² However, little is known about plaque movement during the 4 to 5 days of brachytherapy, which could also contribute to local failure. To address this, we performed intraoperative B-scan ultrasonography at the time of iodine 125 radioactive plaque insertion and removal in 162 consecutive patients with uveal melanoma.

METHODS

PATIENTS

This study was approved by the institutional review board of Washington University. Con-

secutive patients meeting study criteria who were treated between July 1, 2000, and July 31, 2006, were included. Patients were excluded from the study if the ultrasonographic scans were of poor quality in which the plaque margins could not be readily identified ($n = 16$) or if the clinical data were unavailable owing to incomplete or missing records ($n = 19$). A Collaborative Ocular Melanoma Study-style gold plaque was used in all of the cases. Plaque sizes included 12-, 14-, 16-, 18-, 20-, and 22-mm diameters and were chosen to provide at least a 2-mm dosimetric margin around the tumor base. Plaques were initially localized using standard transillumination and indirect ophthalmoscopy with scleral depression. In most cases, a dummy plaque was initially secured to the sclera with temporary 5-0 nylon sutures. Intraoperative B-scan ultrasonography was used to confirm localization of the dummy plaque and readjustments were made as needed. The dummy plaque was then replaced by the active plaque and ultrasonography was performed again. The active plaque was repositioned until optimal localization was achieved. The ultrasonographic scans used for this study were taken after final repositioning of the active plaque. At least three and as many as five 5-0 nylon sutures were used to secure the plaque to the sclera. It was necessary to disinsert a rectus muscle to allow surgical access to the tumor in 68 cases. In these cases, the muscle was secured with a double-locking 5-0 Vicryl

Author Affiliations: Ocular Oncology Service, Department of Ophthalmology and Visual Sciences (Drs Almony, Breit, and Harbour) and Department of Radiation Oncology (Drs Zhao and Mansur and Mr Garcia-Ramirez), Washington University School of Medicine, St Louis, Missouri.

Table 1. Summary of Clinical Findings in 162 Patients

Clinical Factor	Summary Statistics
Female, No. (%)	81 (50)
Right eye, No. (%)	83 (51)
Tumor location, No. (%) ^a	
Temporal	62 (38)
Inferior	50 (30)
Superior	62 (38)
Nasal	46 (28)
Juxtapapillary	48 (30)
Distance to fovea, mean (median; range), mm	3.70 (3.00; 0.00 to 20.00)
Distance to optic disc, mean (median; range), mm	3.79 (3.00; 0.00 to 20.00)
Muscle disinsertion, No. (%)	
Lateral rectus	28 (17)
Inferior rectus	15 (9)
Medial rectus	10 (6)
Superior rectus	16 (10)
Largest basal tumor diameter, mean (median; range), mm	11.15 (11.00; 3.00 to 20.00)
Tumor thickness, mean (median; range), mm	4.74 (4.00; 1.35 to 12.88)
Plaque diameter, No. (%)	
12 mm	8 (5)
14 mm	46 (28)
16 mm	44 (27)
18 mm	29 (18)
20 mm	19 (12)
22 mm	16 (10)
Notched plaque, No. (%)	43 (27)
Plaque tilt at insertion, mean (median; range), mm	0.30 (0.00; 0.00 to 2.75)
Plaque tilt at removal, mean (median; range), mm	1.12 (1.11; 0.00 to 4.74)
Change in plaque tilt, mean (median; range), mm	+0.82 (+0.78; -1.87 to +3.69)

^aSum is greater than 100% because some tumors were scored in 2 locations (eg, superior and nasal).

suture and disinserted, then (after plaque insertion) it was reattached to its insertion by a hang-back suture or secured to the plaque during the course of brachytherapy. In 6 cases, the anterior portion of the inferior oblique muscle insertion was dissected to minimize displacement of an overlying plaque. Despite the use of cautery for hemostasis, a hematoma was often detected at plaque removal. Subsequent to this realization, dissection of the inferior oblique was not performed.

Plaques were designed to deliver a dose of 85 Gy in 96 hours to the prescription point at a dose rate of 0.8 to 0.9 Gy/h. The prescription point was either the tumor apex (for tumors > 5 mm in thickness) or the apex plus 1.0 mm (for tumors ≤ 5 mm), consistent with current guidelines.¹³ Plaque placement and removal were performed in the operating room with local anesthesia (retrobulbar injection of 5 mL of lidocaine and bupivacaine hydrochloride [Marcaine; AstraZeneca, London, England]) in 8 patients and general anesthesia in 154 patients. No differences in plaque positioning or tilt were noted between those patients receiving general vs local anesthesia. Patients were prepared and draped for surgery to remove the plaque; before surgery was initiated, ultrasonography was performed in the same manner as at plaque insertion.

Ultrasonography was performed by one of us (J.W.H.) using diagnostic ophthalmic ultrasonography model System-ABD (Innovative Imaging, Inc, Sacramento, California), with the

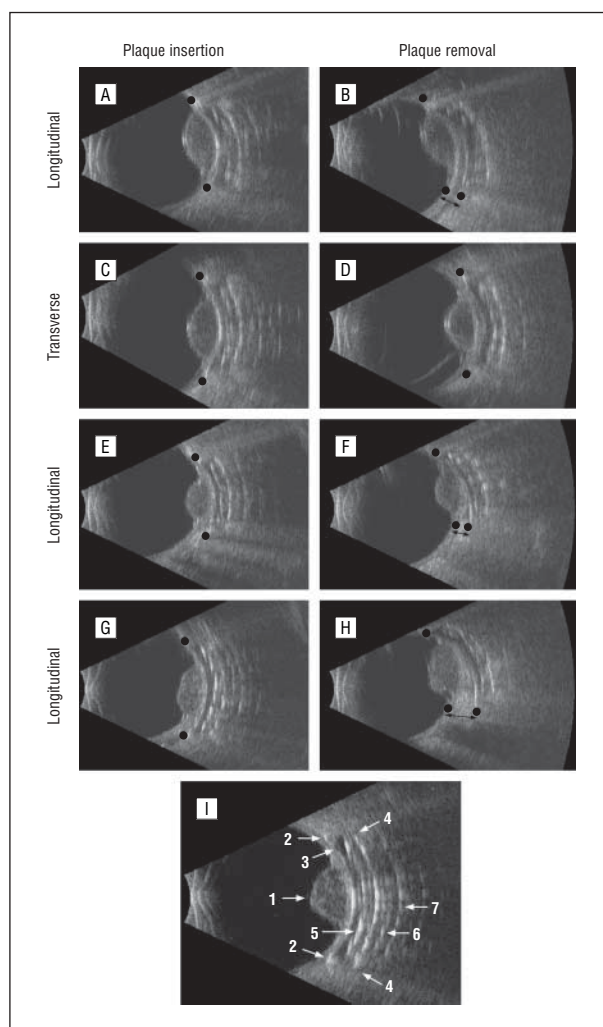


Figure 1. Intraoperative ultrasonography of plaques at insertion and removal. A patient had longitudinal scans showing no tilt at plaque insertion (A) and 1.1 mm of tilt (arrow) at removal (B) and transverse scans showing no horizontal movement at insertion (C) or removal (D). The tilt was due to optic nerve sheath impingement. Another patient had no plaque tilt at insertion (E) and 1.3 mm of tilt (arrow) at removal due to the inferior oblique muscle (F). A third patient had no plaque tilt at insertion (G) and 2.8 mm of tilt (arrow) at removal due to episcleral hematoma (H). I, Example of intraoperative ultrasonographic image to illustrate key structures. 1 indicates tumor; 2, acoustic reflections delineating inner plaque edges; 3, outer scleral surface with no intervening space (or tilt) between the sclera and plaque; 4, outer plaque edge; 5, acoustic reflections delineating radioactive seeds suspended in the silastic insert; 6, acoustic echo of the radioactive seed in the orbit posterior to the plaque; and 7, acoustic echo of the posterior plaque surface.

B-scan probe sheathed in a sterile sleeve as previously described.^{10,14} Longitudinal and transverse ultrasonographic sections were obtained at plaque insertion and removal. Horizontal plaque movement was assessed in the first 106 patients by digitizing and reconstructing images from the longitudinal and transverse scans using ImageJ software (National Institutes of Health, Bethesda, Maryland) to show the plaque position relative to the tumor. Posterior plaque tilt was assessed in all of the 162 patients and was defined as the distance from the outer scleral surface to the inner edge of the plaque measured at the level of the posterior tumor margin. The tilt measurements were obtained directly from the ultrasonography machine at the time of plaque insertion and removal.

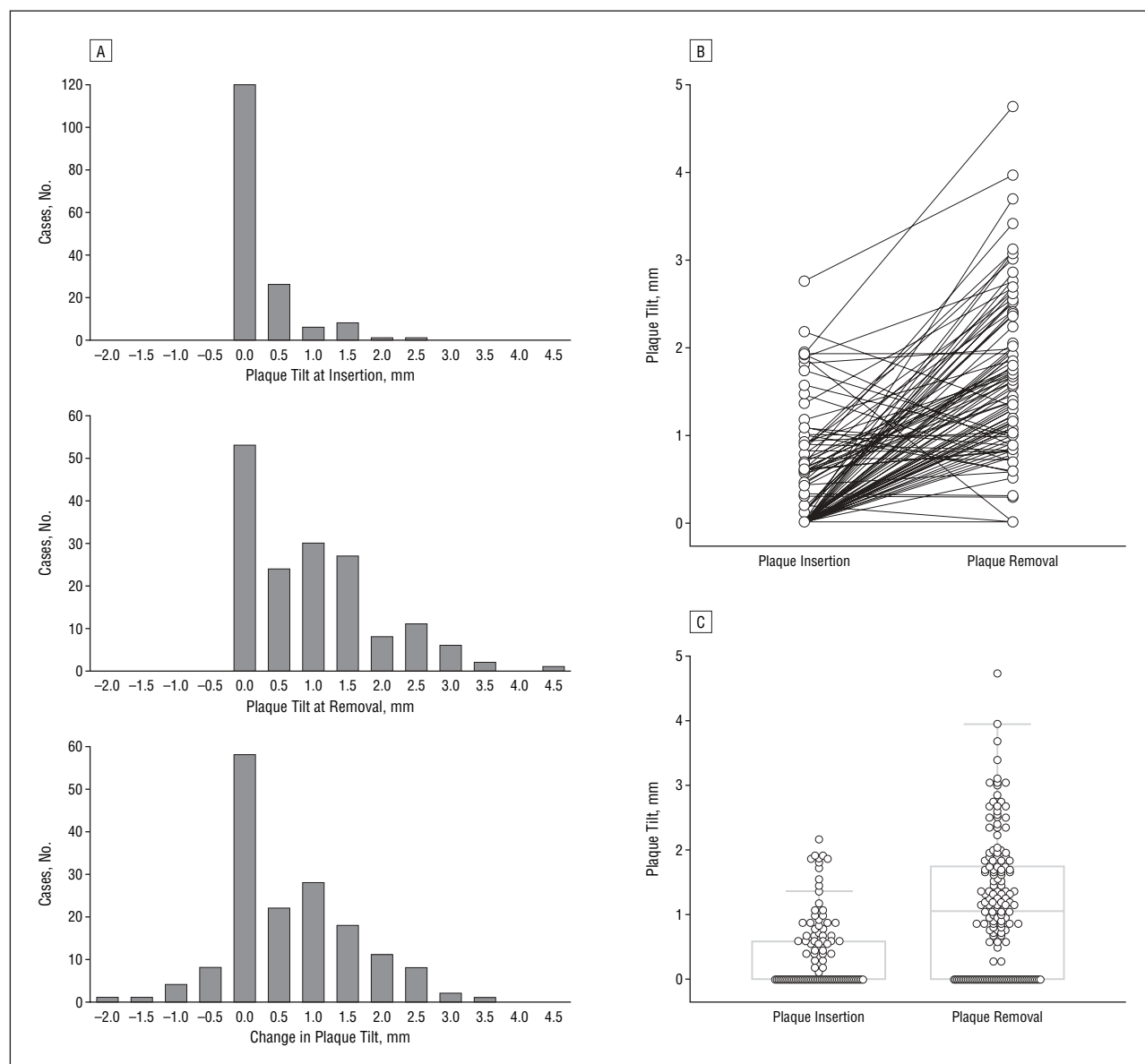


Figure 2. Plaque tilt at insertion and removal. A, Number of cases with plaque tilt at insertion and removal and net change. B, Plaque tilt at insertion and removal for individual cases. C, Box-and-whisker plot of plaque tilt at insertion and removal. The central box represents the values from the 25th to 75th percentiles. The middle line represents the median. The vertical bars extend from the minimum to the maximum values, excluding outliers. $P < .001$ based on the Mann-Whitney test.

STATISTICAL ANALYSIS

Statistical analysis was performed using MedCalc version 9.2.0.2 software (MedCalc Software, Mariakerke, Belgium). The Spearman rank correlation coefficient was used to test the association between continuous variables. The Mann-Whitney test was used to test the significance of the difference between variables based on dichotomous categories.

DOSIMETRIC CALCULATIONS

The brachytherapy module of the Philips treatment planning system (Pinnacle version 7.0; Philips Radiation Oncology Systems, Milpitas, California) was used to determine the radiation dose to the prescription point and the tumor apex. Variables included plaque diameter, prescription depth, plaque tilt at removal, tumor thickness, and dose interval. Plaque tilt at the center point of the plaque was approximated to be half of the tilt measured at the posterior tumor margin. The actual dose

to the prescription point, taking plaque tilt into account, was used to calculate the actual dose to the tumor apex for each case. A graph was generated to estimate the effect of plaque tilt on dose to the prescription point using 2 standard prescription points (5 and 10 mm) and 2 plaque sizes (12 and 20 mm).

RESULTS

HORIZONTAL PLAQUE MOVEMENT ALONG THE SCLERA

Clinical and intraoperative findings are summarized in **Table 1**. Horizontal plaque movement along the sclera was evaluated in the first 106 tumors using longitudinal and transverse ultrasonographic images obtained at the time of plaque insertion and removal (**Figure 1A-D**). Horizontal movement was less than 1.0 mm in 91 pa-

Table 2. Statistical Association Between Clinical Parameters and Plaque Tilt^a

Parameter	P Value for Plaque Tilt at Insertion	P Value for Plaque Tilt at Removal	P Value for Change in Plaque Tilt
Age	.06	.55	.87
Male sex	.95	.009	.005
Right eye	.72	.27	.34
Tumor distance to optic disc	.17	< .001	< .001
Tumor distance to fovea	.01	< .001	< .001
Inferior location	.53	.12	.15
Superior location	.18	.82	.38
Nasal location	.76	.47	.71
Temporal location	.50	.56	.51
Largest tumor diameter	.25	.59	.54
Tumor thickness	.16	.86	.38
Rectus muscle disinserted	.34	.28	.30
Plaque diameter	.42	.20	.20
Notched plaque	.69	.001	.004
Episcleral hematoma at plaque removal	.81	.009	.002
Plaque tilt at insertion	NA	< .001	.08
Plaque tilt at removal	NA	NA	< .001

Abbreviation: NA, not applicable.

^a P values based on Spearman rank correlation coefficient or Mann-Whitney test.

tients (86%). In none of the cases was a tumor margin exposed by horizontal movement. In particular, 2 of the 3 tumors that exhibited local recurrence were analyzed for horizontal plaque movement and in both cases there was less than 2 mm of shift at plaque removal, indicating that the local failure was unlikely to be due to an exposed tumor margin. Because no significant horizontal plaque movement was identified, subsequent patients were analyzed only for plaque tilt.

PLAQUE TILT AWAY FROM THE SCLERA

Plaque tilt away from the sclera was evaluated in 162 patients. In each case, plaque tilt was greatest at the posterior edge of the plaque and was most readily detected on longitudinal ultrasonographic scans (Figure 1E-I). At plaque insertion, no plaque tilt was detected in 109 patients (67%), 0.1 to 1.0 mm of tilt was detected in 38 patients (23%), and tilt greater than 1.0 mm was detected in 15 patients (9%) (Figure 2A). Plaque tilt was detectable by ultrasonography but not indirect ophthalmoscopy or transillumination. In most cases, tilt detected at plaque insertion could be corrected by repositioning the plaque and verifying placement with ultrasonography. At plaque removal, no plaque tilt was detected in 51 patients (31%), 0.1 to 1.0 mm of tilt was detected in 26 patients (16%), 1.1 to 2.0 mm of tilt was detected in 58 patients (36%), and tilt greater than 2.0 mm was detected in 27 patients (17%) (Figure 2A). Although some plaques demonstrated a decrease in tilt between insertion and removal, most showed an increase (Figure 2B). The overall difference in plaque tilt at insertion and removal was highly significant ($P < .001$) (Figure 2C). Tilt at plaque removal was caused by prescleral hematoma in 7 patients, inferior ob-

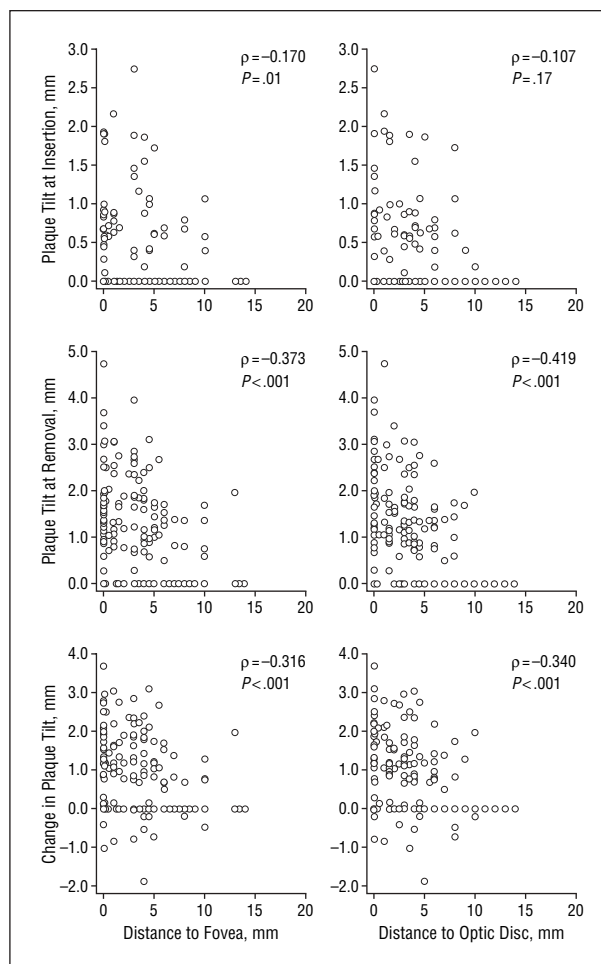


Figure 3. Scatterplots showing associations between plaque tilt (at insertion, removal, and net change) and tumor distance to the fovea and optic disc. Spearman ρ and P values are based on rank correlation.

lique muscle in 4 patients, optic nerve sheath or posterior ciliary nerves and vessels in 34 patients, and undetermined in the remainder of patients. Only 3 local treatment failures occurred in this study. These cases showed no tilt at plaque insertion and 1.95 mm, 2.23 mm, and 3.05 mm of tilt at plaque removal. Statistical analysis was performed to identify risk factors for plaque tilt (Table 2). The only factor associated with plaque tilt at the time of insertion was decreased distance of the tumor margin to the fovea ($P = .01$) (Figure 3). Factors associated with plaque tilt at removal included male sex ($P = .009$), decreased tumor distance to the fovea and optic disc ($P < .001$ for both), notched plaque ($P = .001$), and episcleral hematoma underlying the plaque at the time of plaque removal ($P = .009$) (Figure 3 and Figure 4).

DOSIMETRIC CALCULATIONS

The actual radiation dose to the tumor apex was calculated for each patient based on the amount of tilt at plaque removal and the assumption that tilt occurred at a constant rate. Plaque tilt resulted in a reduction in the actual radiation dose to the tumor apex by more than 10% in 37 patients (23%) and more than 20% in 6 patients

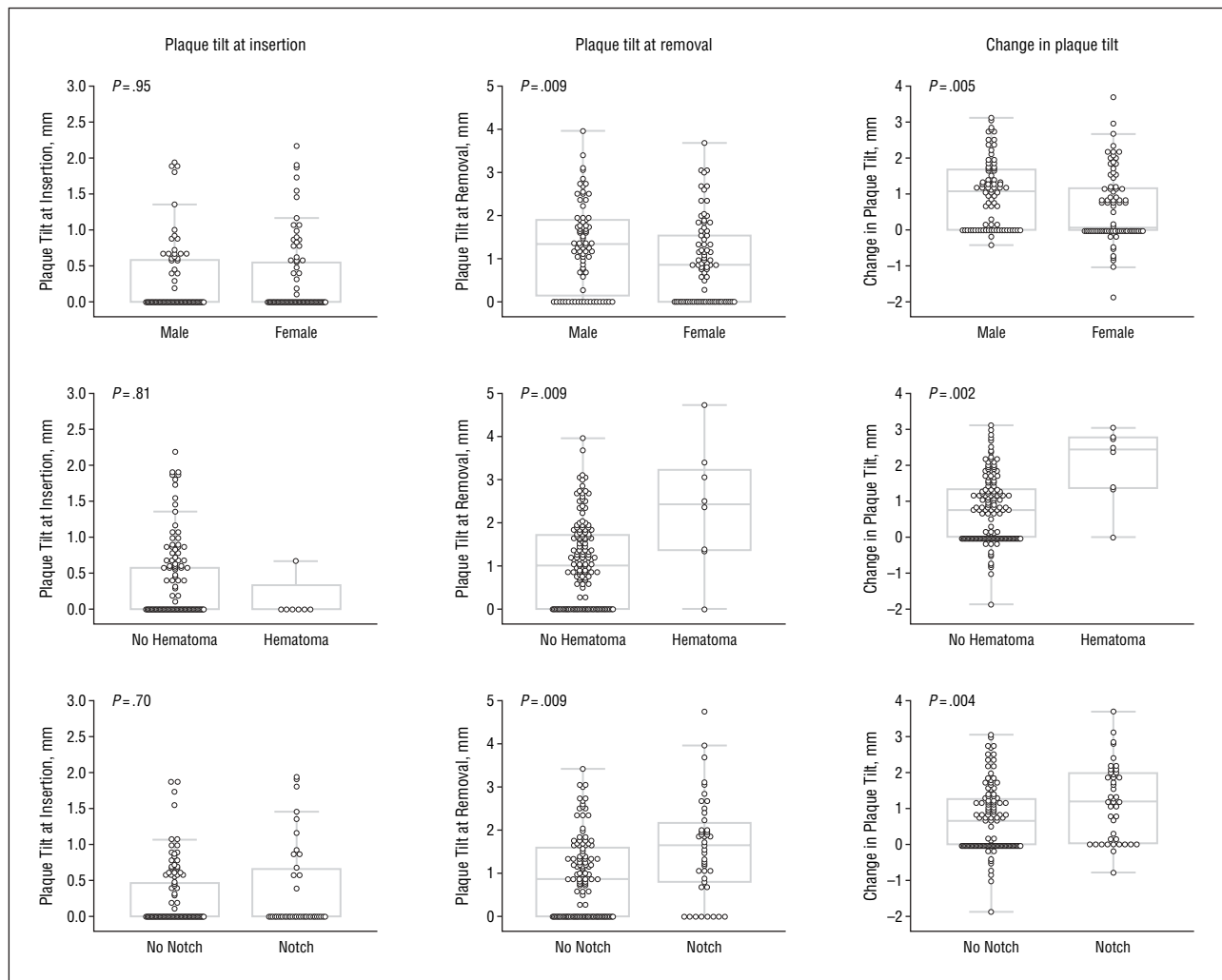


Figure 4. Association between plaque tilt (insertion, removal, and change) and sex, episcleral hematoma, and notched plaque. Boxes represent the 25th to 75th percentiles. Middle lines represent the median values. The vertical lines extend from minimum to maximum values.

(4%). Reduction in the radiation dose to the prescription point caused by plaque tilt was greater for tumors with increased basal diameter and decreased thickness ($P < .001$ for each). The estimated effect of plaque tilt on the dose to the prescription point was graphed using 2 standard prescription points (5 and 10 mm) and 2 plaque sizes (12 and 20 mm) (**Figure 5**).

COMMENT

This study confirmed that most plaques can be accurately localized using intraoperative ultrasonography at plaque insertion. However, a surprisingly large proportion of plaques became tilted during the 4 to 5 days that the plaque was in place as demonstrated by intraoperative ultrasonography at plaque removal. Plaque tilt greater than 1.0 mm was detected in fewer than 10% of the plaques at the time of insertion but in more than half of the plaques at plaque removal. Tilt was greatest at the posterior edge of the plaque away from the sutures that secure the plaque to the sclera. Collaborative Ocular Melanoma Study-style plaques typically have 6 eyelets that span almost 180° of the plaque, so use of the more pe-

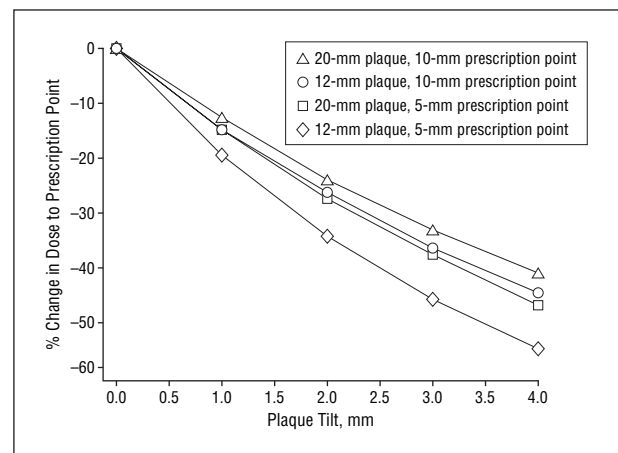


Figure 5. Effect of plaque tilt on dose to the prescription point using 2 standard prescription points (5 and 10 mm) and 2 plaque sizes (12 and 20 mm).

ripheral eyelets may help to reduce tilt. Other plaque styles that incorporate fewer eyelets that are concentrated toward one end of the plaque and very thin plaques such as those used for ruthenium 106 brachytherapy may be

at even higher risk of tilt. There were not enough local treatment failures to determine statistically whether tilt was associated with local failure, but it is perhaps significant that all of the 3 local failures had no tilt at plaque insertion and 1.95 to 3.05 mm of tilt at plaque removal, which was greater than the mean tilt at plaque removal (mean, 1.12 mm). Because some patients were followed up for a relatively short time, longer follow-up will be needed to determine whether there will be more local failures in the future and whether there will be an association between plaque tilt and patient survival.

The risk for plaque tilt was greatest for posterior tumors near the fovea and optic disc, which are well known to have a higher local failure rate.^{2,4,6,9,15} Notched plaques were used for juxtapapillary tumors in which the posterior tumor margin was located less than 1.5 mm from the disc, thus explaining the association between notched plaques and plaque tilt. Plaque localization is more difficult for tumors located posteriorly near the disc and fovea, where surgical access and visualization are constrained by the orbital anatomy. In addition, it can be difficult to achieve close apposition of the plaque to the sclera owing to obstruction by the optic nerve sheath, inferior oblique muscle, and posterior ciliary vessels and nerves.^{10-12,16} These difficulties could explain the higher failure rate for posterior tumors in centers where ultrasonographic localization is not routinely performed.^{2,4,6,9,15} However, we show in this study that posterior location is a risk factor for plaque tilt (and possibly local failure) even when intraoperative ultrasonography at plaque insertion has confirmed correct plaque localization. The same anatomical structures that hamper accurate plaque localization can become congested and edematous while the plaque is in place, displacing the plaque from the scleral surface. Consequently, we now routinely perform intraoperative ultrasonography at plaque removal, and if significant tilt is observed, the plaque can be left in longer or adjuvant transpupillary thermotherapy can be performed.

The risk of plaque tilt for posterior tumors is compounded by the fact that posterior tumors are often thin and small, and the reduction in radiation dose to the tumor per millimeter of plaque tilt is greater for thinner tumors and smaller plaque diameters. Consequently, we routinely define the prescription point 1.0 mm deeper than the actual tumor thickness for posteriorly located tumors thinner than 5 mm.

An infrequent but highly significant risk factor for plaque tilt was episcleral hematoma between the plaque and sclera at the time of plaque removal. Indeed, episcleral hematoma was responsible for many of the more extreme cases of tilt (Figure 1). There was no association with excessive bleeding at plaque insertion or anticoagulant use in these cases. However, a rectus muscle had been disinserted or the inferior oblique muscle had been dissected in many of the cases with hematoma-associated plaque tilt. Therefore, we now avoid cutting extraocular muscles whenever possible. The association between plaque tilt and male sex is not clear. Perhaps males are more likely to develop swelling of the episcleral tissues while the plaque is in place.

Because plaque tilt cannot be detected by indirect ophthalmoscopy, transillumination, lighted depressors, and other commonly used localization techniques, we use intraoperative ultrasonography at plaque insertion to localize plaques and at plaque removal to identify any significant plaque tilt. If plaque tilt is identified at removal, we perform postoperative adjuvant transpupillary thermotherapy. A prospective study is ongoing to evaluate whether these guidelines will have a significant impact on local control, metastasis, and survival.

Submitted for Publication: April 13, 2007; final revision received June 8, 2007; accepted June 11, 2007.

Correspondence: J. William Harbour, MD, Ocular Oncology Service, Department of Ophthalmology and Visual Sciences, Washington University School of Medicine, 660 S Euclid Ave, Box 8096, St Louis, MO 63110 (harbour@vision.wustl.edu).

Financial Disclosure: None reported.

REFERENCES

1. Harbour JW. Clinical overview of uveal melanoma: introduction to tumors of the eye. In: Albert DM, Polans A, eds. *Ocular Oncology*. New York, NY: Marcel Dekker; 2003:1-18.
2. Char DH, Quivey JM, Castro JR, Kroll S, Phillips T. Helium ions vs iodine 125 brachytherapy in the management of uveal melanoma: a prospective, randomized, dynamically balanced trial. *Ophthalmology*. 1993;100(10):1547-1554.
3. Garretson BR, Robertson DM, Earle JD. Choroidal melanoma treatment with iodine 125 brachytherapy. *Arch Ophthalmol*. 1987;105(10):1394-1397.
4. Karlsson UL, Augsburger JJ, Shields JA, Markoe AM, Brady LW, Woodleigh R. Recurrence of posterior uveal melanoma after 60Co episcleral plaque therapy. *Ophthalmology*. 1989;96(3):382-388.
5. Lommatzsch PK. Beta-irradiation of choroidal melanoma with 106Ru/106Rh applicators: 16 years' experience. *Arch Ophthalmol*. 1983;101(5):713-717.
6. Quivey JM, Augsburger J, Snelling L, Brady LW. 125I plaque therapy for uveal melanoma: analysis of the impact of time and dose factors on local control. *Cancer*. 1996;77(11):2356-2362.
7. Vrabec TR, Augsburger JJ, Gamel JW, Brady LW, Hernandez C, Woodleigh R. Impact of local tumor relapse on patient survival after cobalt 60 plaque radiotherapy. *Ophthalmology*. 1991;98(6):984-988.
8. Gragoudas ES, Egan KM, Seddon JM, Walsh SM, Munzenrider JE. Intraocular recurrence of uveal melanoma after proton beam irradiation. *Ophthalmology*. 1992;99(5):760-766.
9. De Potter P, Shields CL, Shields JA, Cater JR, Brady LW. Plaque radiotherapy for juxtapapillary choroidal melanoma: visual acuity and survival outcome. *Arch Ophthalmol*. 1996;114(11):1357-1365.
10. Harbour JW, Murray TG, Byrne SF, et al. Intraoperative echographic localization of iodine 125 episcleral radioactive plaques for posterior uveal melanoma. *Retina*. 1996;16(2):129-134.
11. Tabandeh H, Chaudhry NA, Murray TG, et al. Intraoperative echographic localization of iodine-125 episcleral plaque for brachytherapy of choroidal melanoma. *Am J Ophthalmol*. 2000;129(2):199-204.
12. Finger PT, Romero JM, Rosen RB, Iezzi R, Emery R, Berson A. Three-dimensional ultrasonography of choroidal melanoma: localization of radioactive eye plaques. *Arch Ophthalmol*. 1998;116(3):305-312.
13. Nag S, Quivey JM, Earle JD, Followill D, Fontanesi J, Finger PT; American Brachytherapy Society. The American Brachytherapy Society recommendations for brachytherapy of uveal melanomas. *Int J Radiat Oncol Biol Phys*. 2003;56(2):544-555.
14. Byrne SF, Green RL. *Ultrasound of the Eye and Orbit*. St Louis, MO: Mosby-Year Book; 1992.
15. Quivey JM, Char DH, Phillips TL, Weaver KA, Castro JR, Kroll SM. High intensity 125-iodine (125I) plaque treatment of uveal melanoma. *Int J Radiat Oncol Biol Phys*. 1993;26(4):613-618.
16. Pavlin CJ, Japp B, Simpson ER, McGowan HD, Fitzpatrick PJ. Ultrasound determination of the relationship of radioactive plaques to the base of choroidal melanomas. *Ophthalmology*. 1989;96(4):538-542.